Object Detection Metrics & Post-processing

IoU, NMS, and mAP

Goals

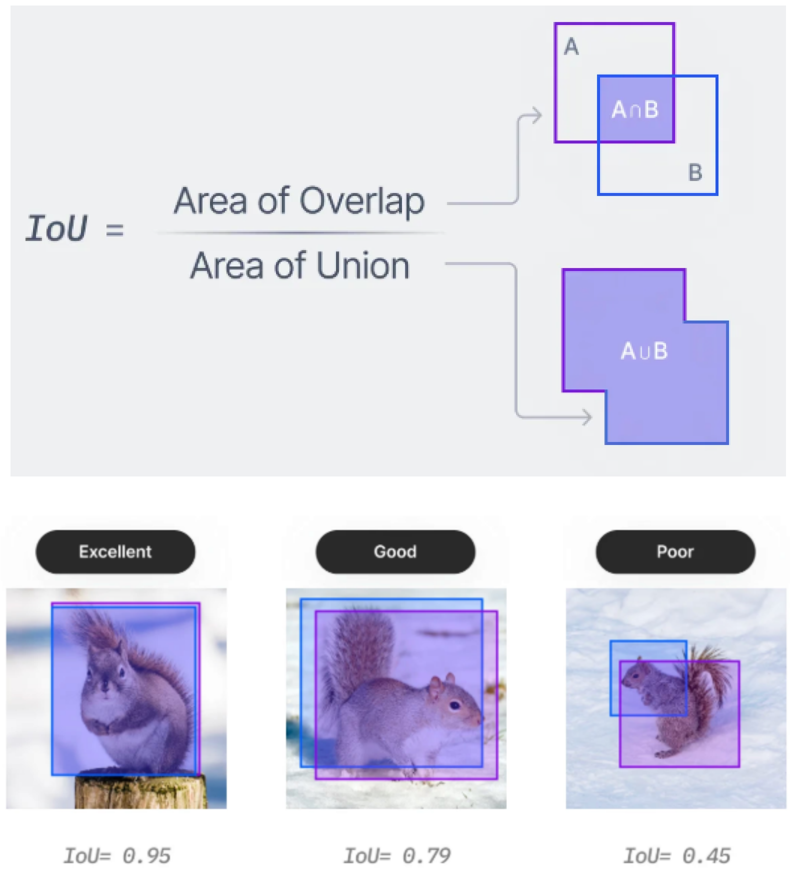
- Understand how detections match to ground truth via IoU
- Know how overlapping predictions are pruned with NMS
- Interpret precision, recall, AP, and mAP

Detector Outputs Refresher

- For each image returns: boxes (x1,y1,x2,y2), class scores, confidence/objectness
- Multiple detections may target the same object → need post-processing
- Evaluation → ties predictions to ground truth using IoU + class match ie. (IoU >threshold and correct class)

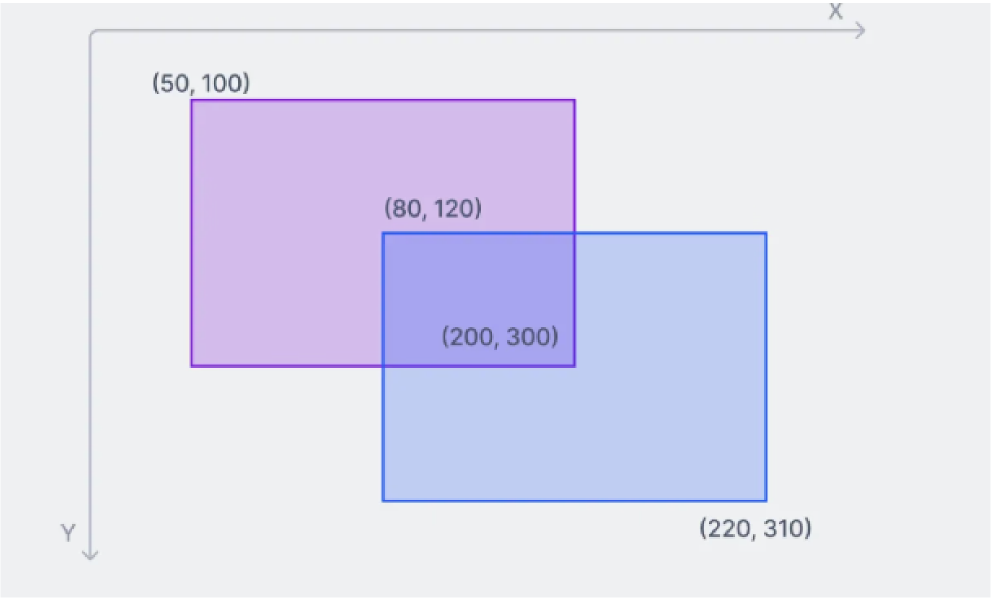
Intersection over Union (IoU)

- Measures quality between predicted and ground-truth boxes
- Range [0,1]; higher is better
- Used for matching: TP if class is correct AND IoU ≥ threshold (e.g., 0.5)



See v7labs tutorial

IoU example



Intersection=(200 – 80) x (300 - 120) = 21,600
Purple= (200 - 50) x (300 - 100) = 30,000
Blue= (220 – 80) x (310 – 120) = 26,600

I0U = 21,600/56,600 = 0.62

v7labs tutorial

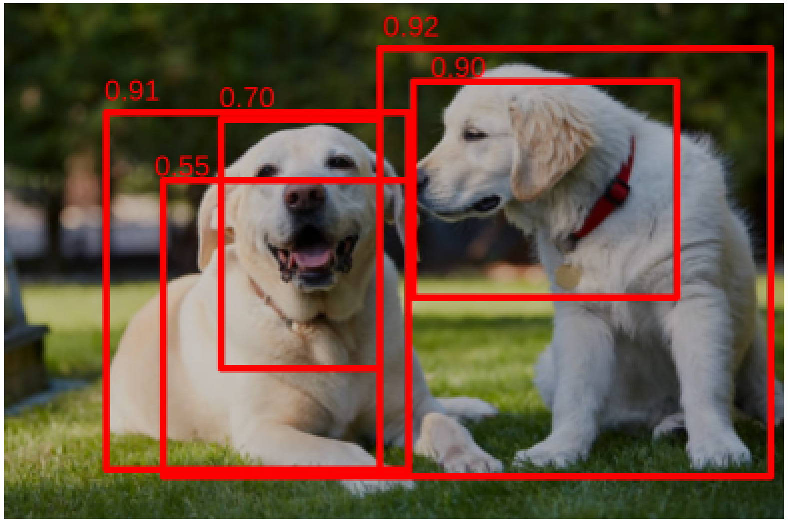
Non-Max Suppression (NMS)

- Why: detectors output many overlapping boxes per object
- NMS: sort boxes by score, keep best, drop overlaps above IoU threshold

NMS example

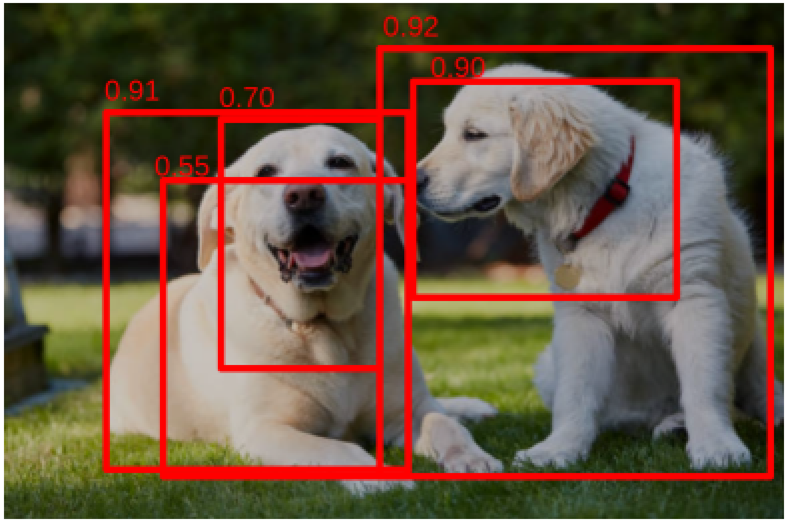
For this Image

NMS example



Object detector proposed these bounding boxes and confidences.

NMS example



Algorithm

- Save highest confidence box
- Compute IoU with all other boxes
- Delete bounding boxes with IoU >threshold (say 0.2)

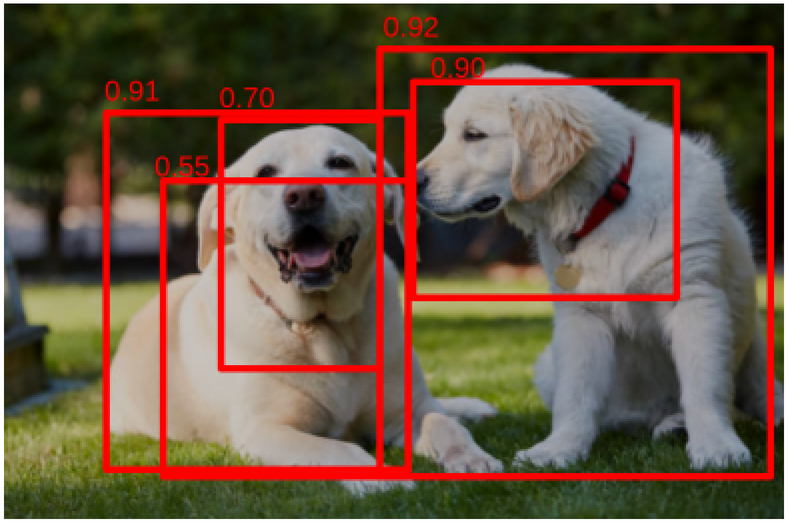
Proposals

Box #	Score	IoU
1	.91	
2	.70	
3	.55	
4	.92	
5	.90	

Keepers

Box #	Score

NMS example



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Box #	Score	IoU
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2	.70	
3	.55	
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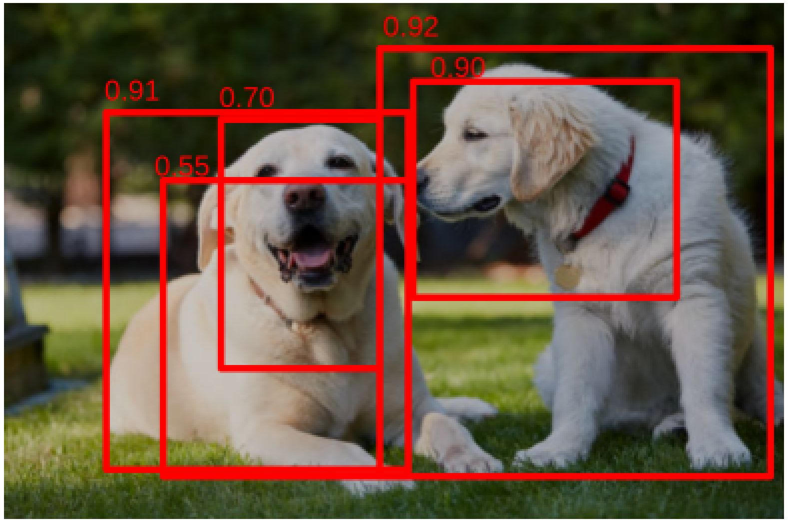
Keepers

Box #	Score
4	.92

NMS example



Algorithm

- Save highest confidence box
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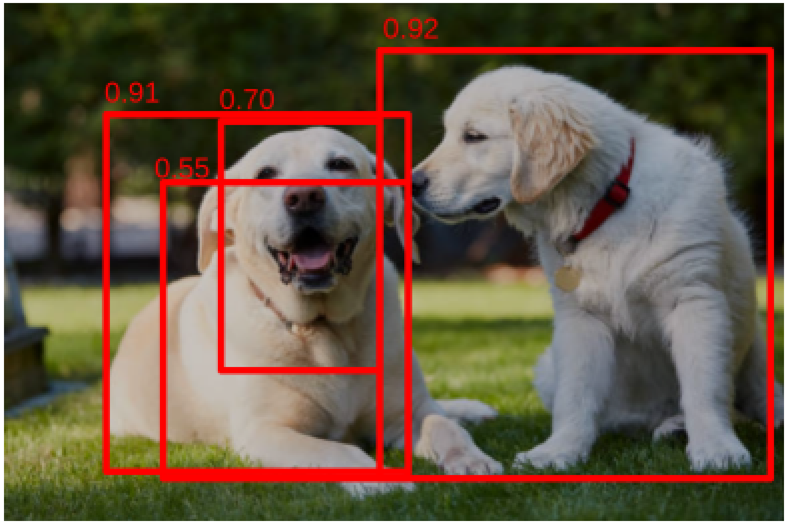
Proposals

Box #	Score	IoU
1	.91	0.1
2	.70	0
3	.55	0.07
5	.90	0.3

Keepers

Box #	Score
4	.92

NMS example



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Keepers

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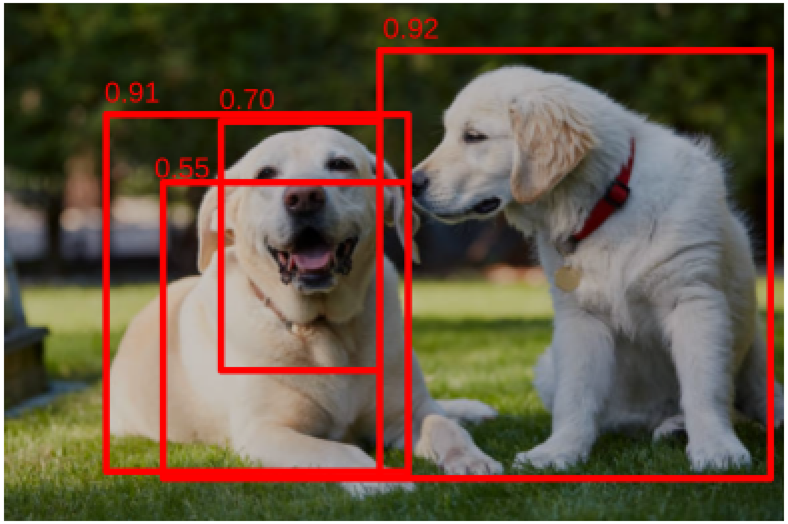
NMS example



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Proposals

Box #	Score
2	.70
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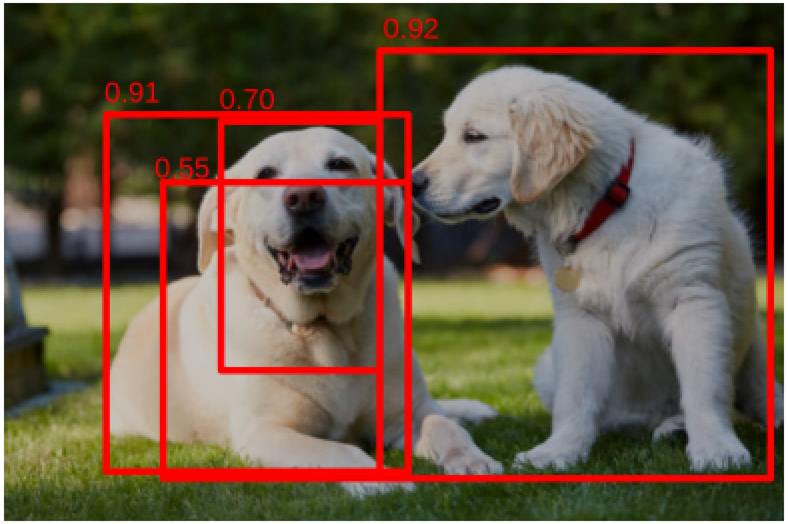
IoU

Keepers

Box #	Score
4	.92
1	.91

NMS example



Algorithm

- Save highest confidence box
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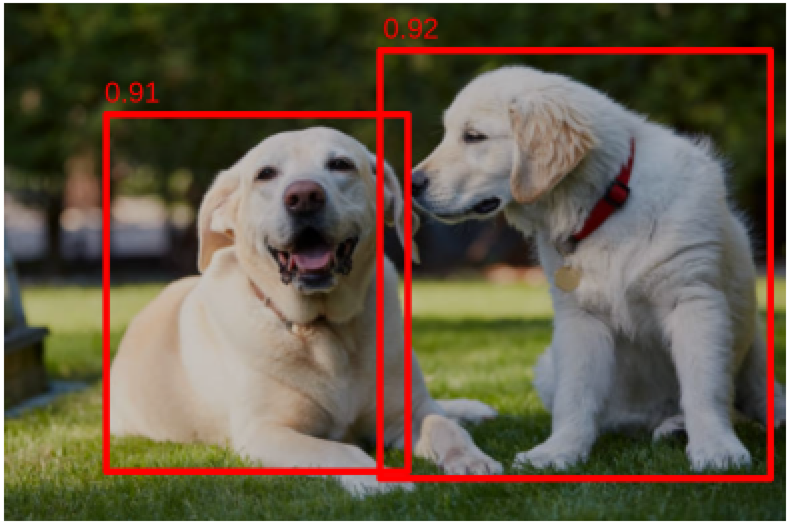
Proposals

Box #	Score	IoU
2	.70	.3
3	.55	.7

Keepers

Box #	Score
4	.92
1	.91

NMS example



Algorithm

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Proposals

Box #	Score

IoU

Keepers

Box #	Score
4	.92
1	.91

Precision, Recall

- Precision = TP/(TP+FP) → of what you predicted, how much was correct?
- Recall = TP/(TP+FN) → of what exists, how much did you find?

Average Precision (AP)

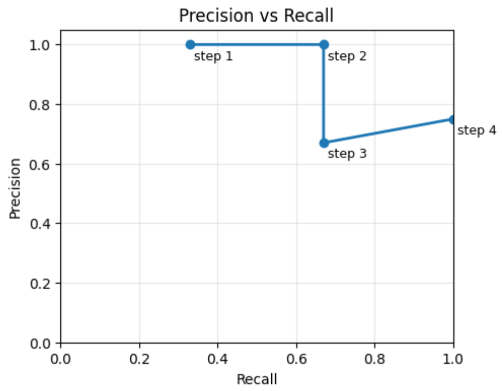
- Measures how good a detector is for **one class** by summarizing its whole precision–recall trade-off into a single number. Fix an IoU threshold (e.g., 0.5), sweep the score/confidence threshold from high→low, plot precision vs. recall, and take the area under that curve. Bigger area ⇒ better detector for that class.

Average Precision (AP)

$P = \frac{T_p}{T_p + F_p}$ $R = \frac{T_p}{T_p + F_n}$

- Ex. Suppose Class A has **3 ground-truth objects**. And you are using a IoU threshold of .5. As you lower the model score threshold (confidence) you accumulate predictions:

step	TP	FP	Precision	Recall
1	1	0	1.00	0.33
2	2	0	1.00	0.67
3	2	1	0.67	0.67
4	3	1	0.75	1.00



Take the area under the curve of plot (max of 1x1=1).Say it comes out to 0.83

That means for this class you have an AP@0.5 = .83

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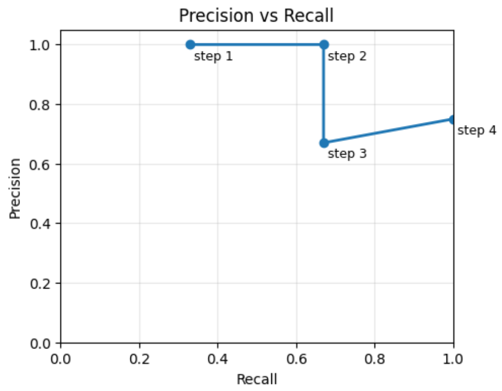
2	2	0	1.00	0.67
3	2	1	0.67	0.67
4	3	1	0.75	1.00

Got 1,
missed 2



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mAP

mAP stands for **Mean Average Precision** and is the most common metric used to evaluate the performance of object detection models in computer vision. It provides a single number that summarizes a model's performance across all object classes and at different levels of detection quality.

A high mAP score indicates the model is good at two things:

1. **Correctly identifying** the object's class (High **Precision**).
2. **Accurately locating** the object in the image (High **Recall** and accurate bounding box).

mAP

Ex. Two classes – calculate mAP@0.5

Class A AP@0.5 = 0.83 (from previous slide)

Class B (different predictions/ground truths) AP@0.5 = 0.60

Then mAP@0.5 across these two classes is the mean:

$$\text{mAP@0.5} = \frac{0.83 + 0.60}{2} = 0.715$$